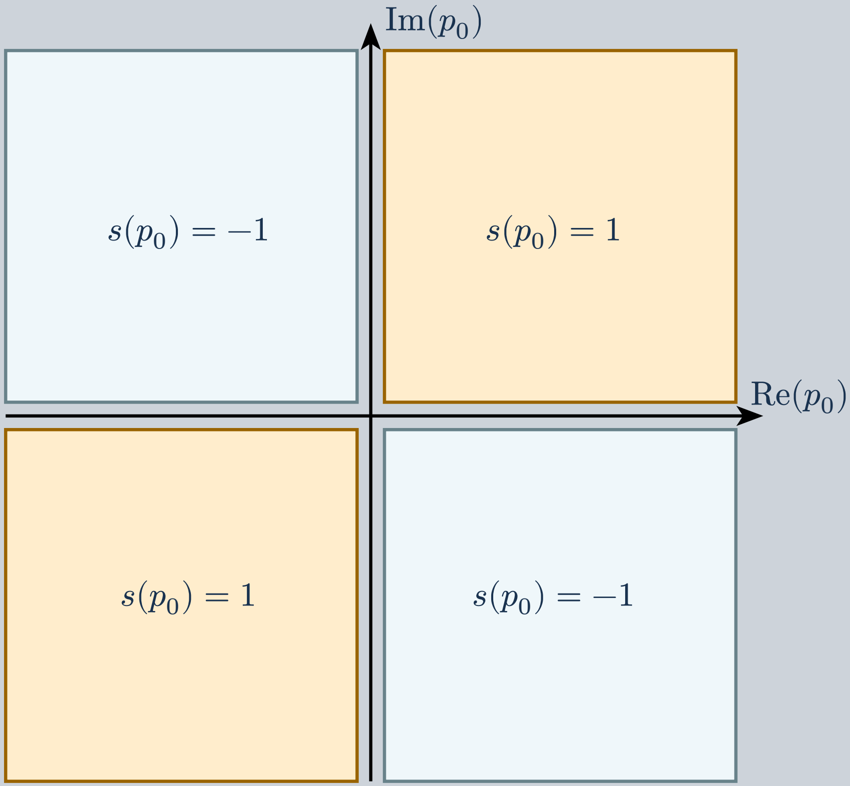
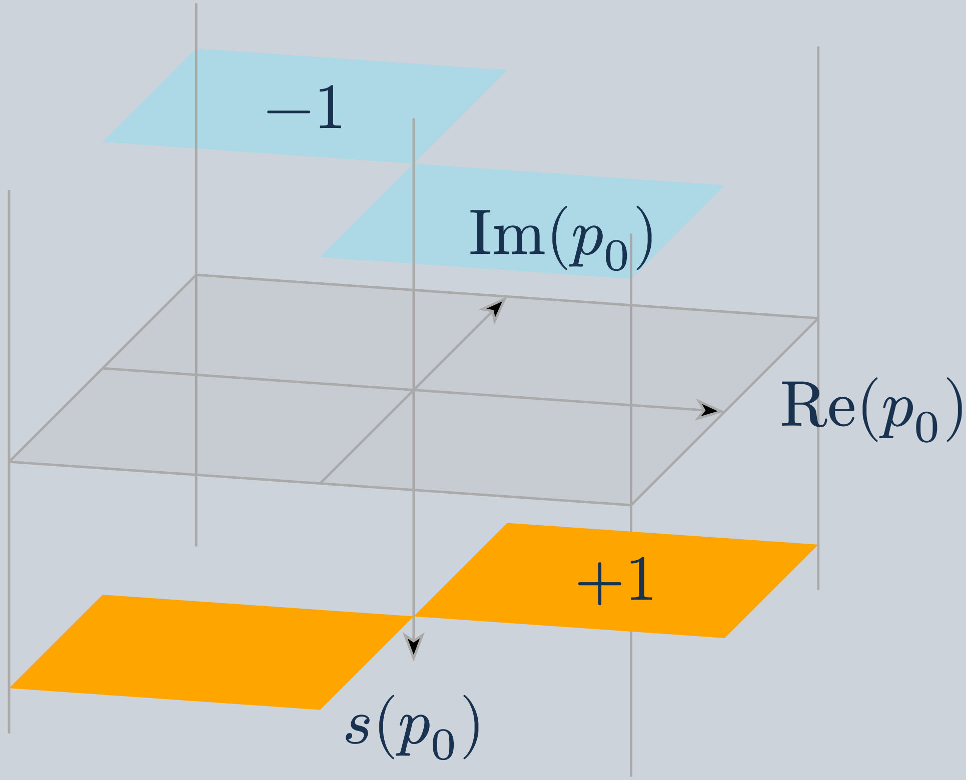
One quadrant-wise sign rule, viewed from above and as a lifted surface. This is the function used here in contour bookkeeping, not the complex phase z/|z|.

1 Locate the quadrant



Equal signs of the real and imaginary parts give $+1$; opposite signs give -1. The sign flips on crossing either axis.

2 Lift the value to a height



The same two values become horizontal sheets. Height encodes the sign; the surface is discontinuous at the axes.

$s(z) = \mathrm{sign}(\mathrm{Re}(z)\,\mathrm{Im}(z))$. These panels show values **away from the axes**. At an axis, a contour calculation must specify the limiting side or its convention.